CBL
Endocrinology Module

By

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Case

A 40 year old male came in outpatient department to see the physician. His complains are:

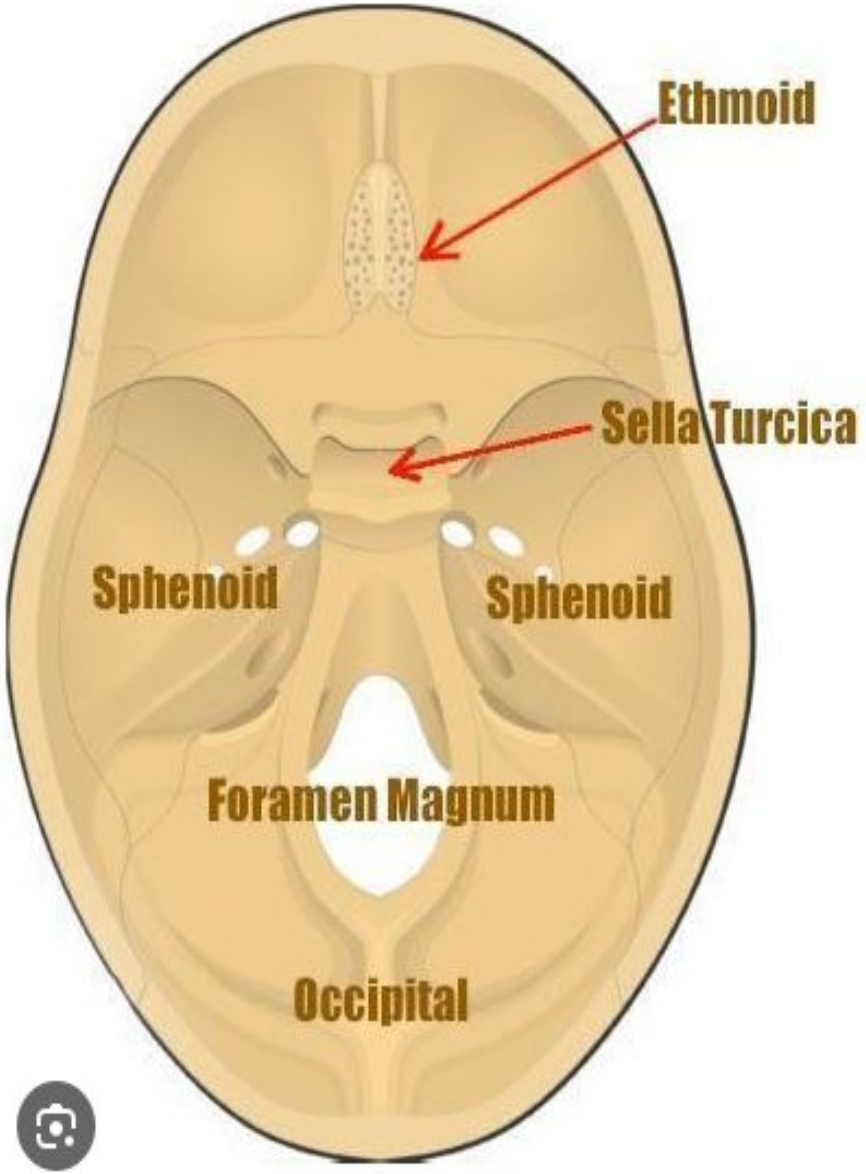
- He recently noticed increase in size of hands and feet, excessive sweating and headache
- He also had difficulty in standing from sitting position

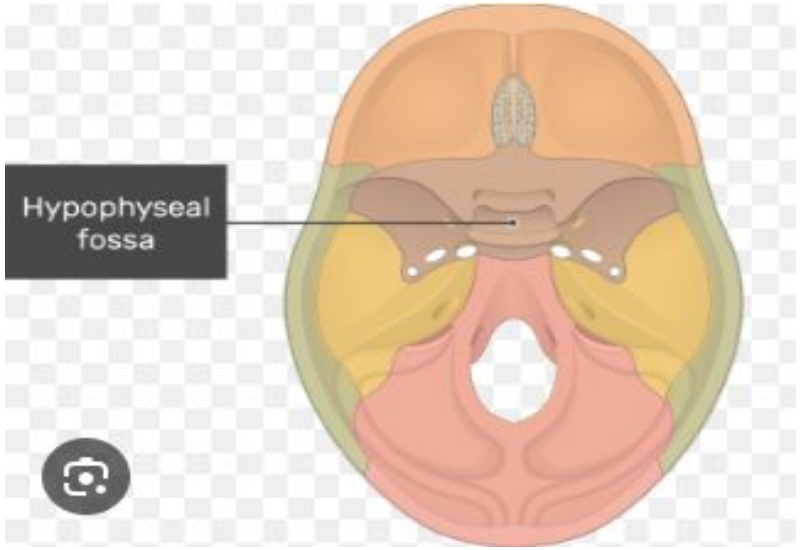
On Examination

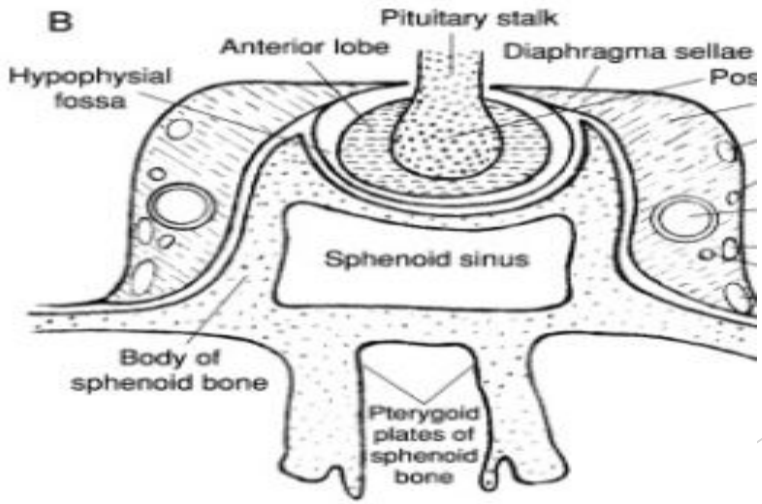
- He is tall, his hands are large
- He has difficulty in standing from sitting position
- He has large tongue, nose and jaw
- He has bilateral temporal visual field defects
- Blood pressure: 160/110 mm Hg

Investigations

- Sugar fasting: 130mg/dL
- WBC: 11000 per micro liter
- PLT: 250 per micro liter
- Hb: 13gm /L
- Serum Prolactin: Raised
- Lateral skull X-ray is abnormal: Enlarge shadow of hypophyseal fossa


Ethmoid
Sella Turcica
Sphenoid
Sphenoid
Foramen Magnum
Occipital


Hypophyseal fossa


B
Pituitary stalk
Anterior lobe
Diaphragma sellae
Hypophysial fossa
Sphenoid sinus
Body of sphenoid bone
Pterygoid plates of sphenoid bone

Question 1

How would you explain the physical signs?

Answer 1

- Middle aged tall man with large hands, nose, jaw and tongue having difficulty in standing from sitting position
- He has bilateral temporal visual field defects
- Blood pressure: 160/110 mm Hg

Question 2

- Which underlying hormonal disorder is he suffering from?

Answer 2

- Growth hormone disorder
- Acromegaly is a hormonal disorder characterized by Growth hormone hyper secretion, predominantly caused by pituitary adenoma

Question 3

Why is he tall?

Answer 3

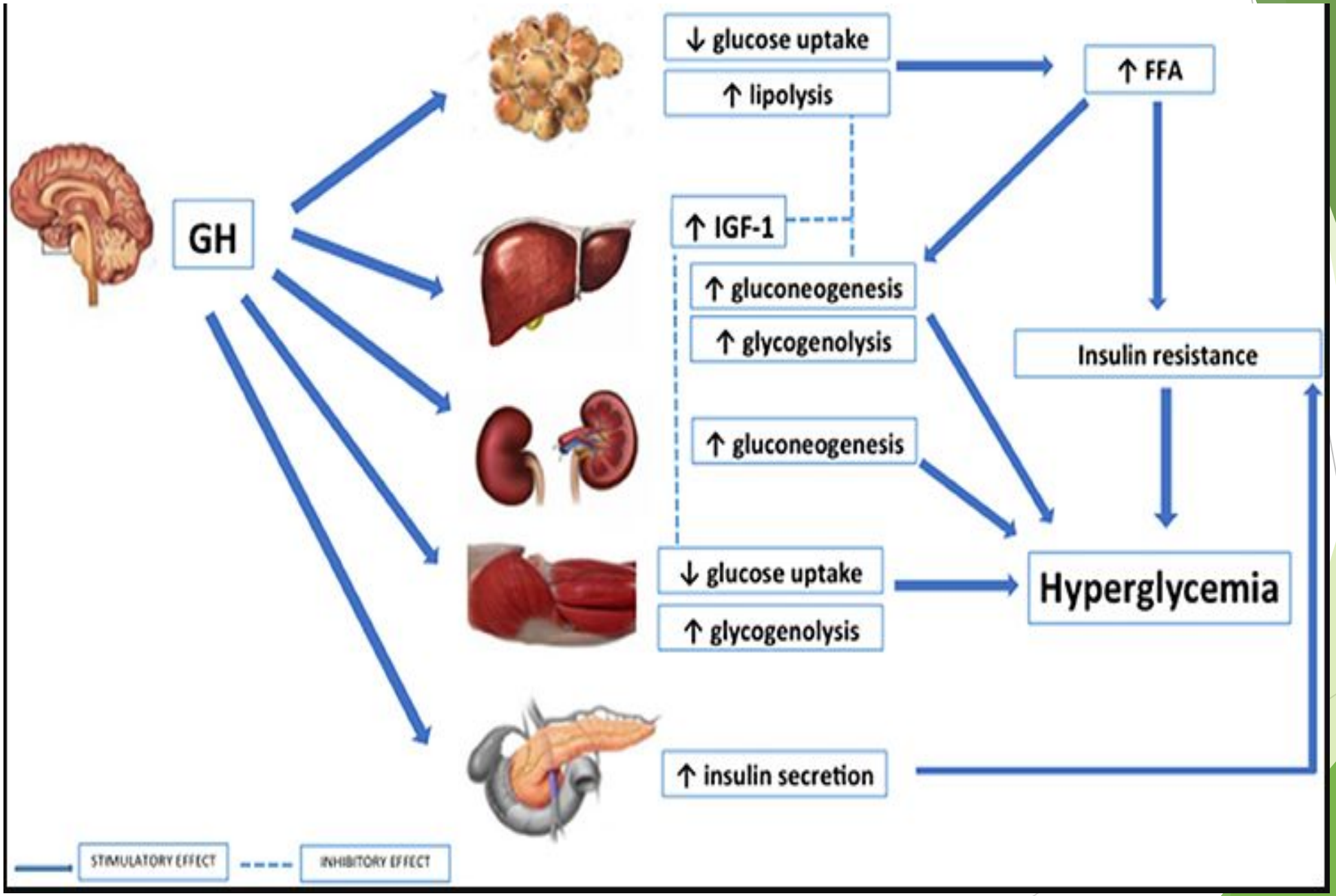
- Acromegaly is a hormonal disorder characterized by Growth hormone hyper secretion, predominantly caused by pituitary adenoma
- Hyper secretion of Growth hormone causes increase in bone size
- Physical changes occur slowly over many years
- Average age at diagnosis is 40-50 years

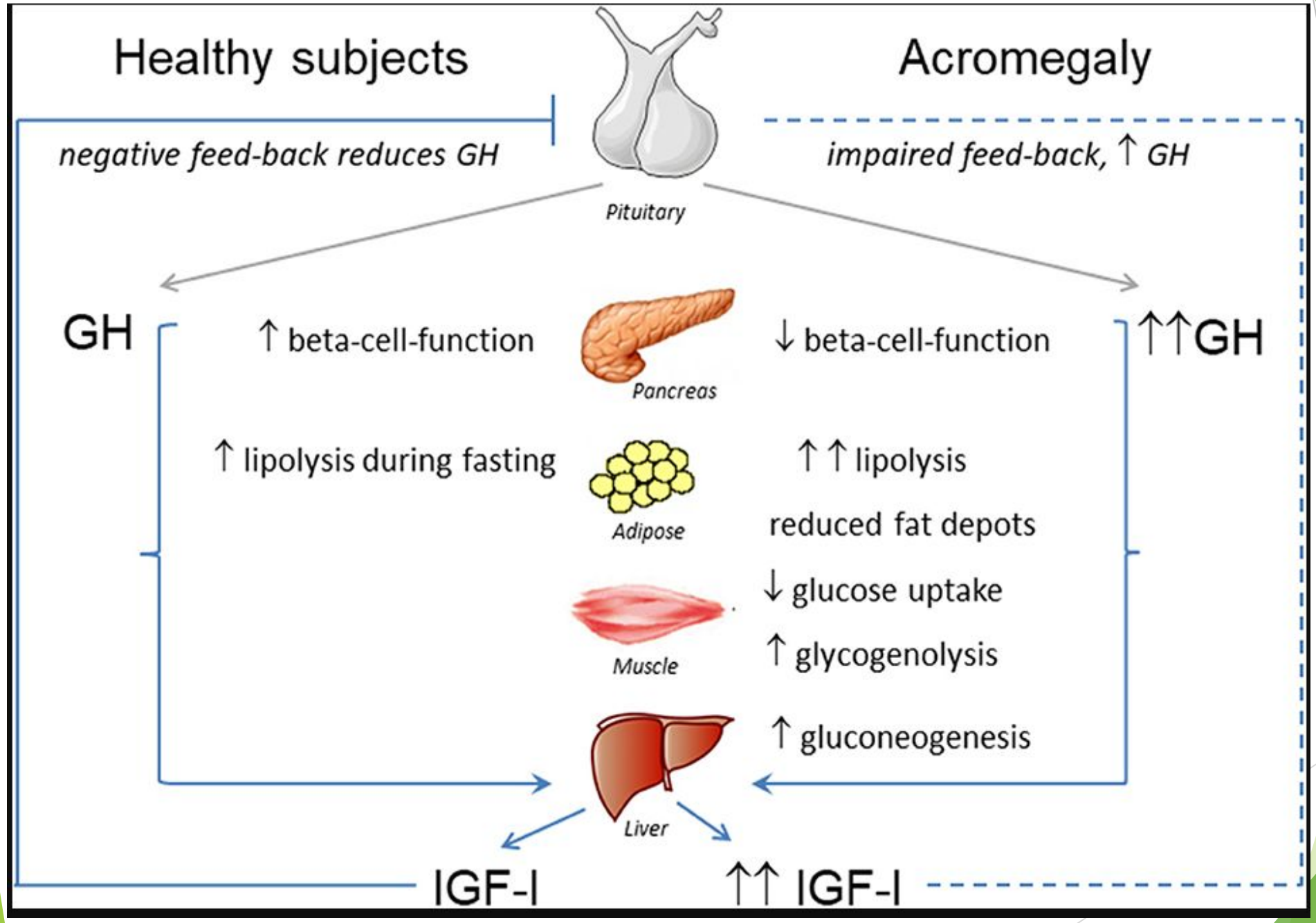
Question 4

Why fasting blood sugar is raised?

Answer 4

- Acromegaly is an uncommon secondary cause of diabetes
- Acromegaly is characterized by excess Growth hormone production
- Excess GH stimulates gluconeogenesis and lipolysis, causing hyperglycemia and elevated free fatty acids levels
- This leads to both hepatic and peripheral insulin resistance, with compensatory hyperinsulinemia


GH
↓ glucose uptake
↑ lipolysis
↑ FFA
↑ IGF-1
↑ gluconeogenesis
↑ glycogenolysis
Insulin resistance
↑ gluconeogenesis
↓ glucose uptake
↑ glycogenolysis
Hyperglycemia
↑ insulin secretion
STIMULATORY EFFECT
INHIBITORY EFFECT

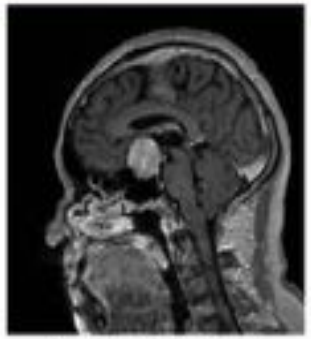

Healthy subjects
Acromegaly
negative feed-back reduces GH
impaired feed-back, ↑ GH
Pituitary
GH
↑ beta-cell-function
↓ beta-cell-function
↑↑GH
Pancreas
↑ lipolysis during fasting
↑↑ lipolysis
Adipose
reduced fat depots
↓ glucose uptake
Muscle
↑ glycogenolysis
↑ gluconeogenesis
Liver
IGF-I
↑↑ IGF-I

Question 5

Comment on blood pressure?

Answer 5

- Hypertension is an important complication of acromegaly, contributing to increase morbidity and mortality
- The excess GH in acromegaly causes insulin resistance and hyperinsulinemia which is associated with enhanced sodium- and water retention, expansion of plasma volume and increased systemic vascular resistance
- Insulin resistance also cause endothelial proliferation that causes increase in vascular resistance



Acromegaly

GH increase

Insulin resistance

Endothelial dysfunction

Hyperinsulinemia

Activation of the renin-angiotensin-aldosterone system

Adrenergic activity increase

Vascular resistance increase

Hypertension

Aldosterone secretion

Water and sodium retention

Blood volume expansion

Question 6

How would you relate increase prolactin level in this patient?

Answer 6

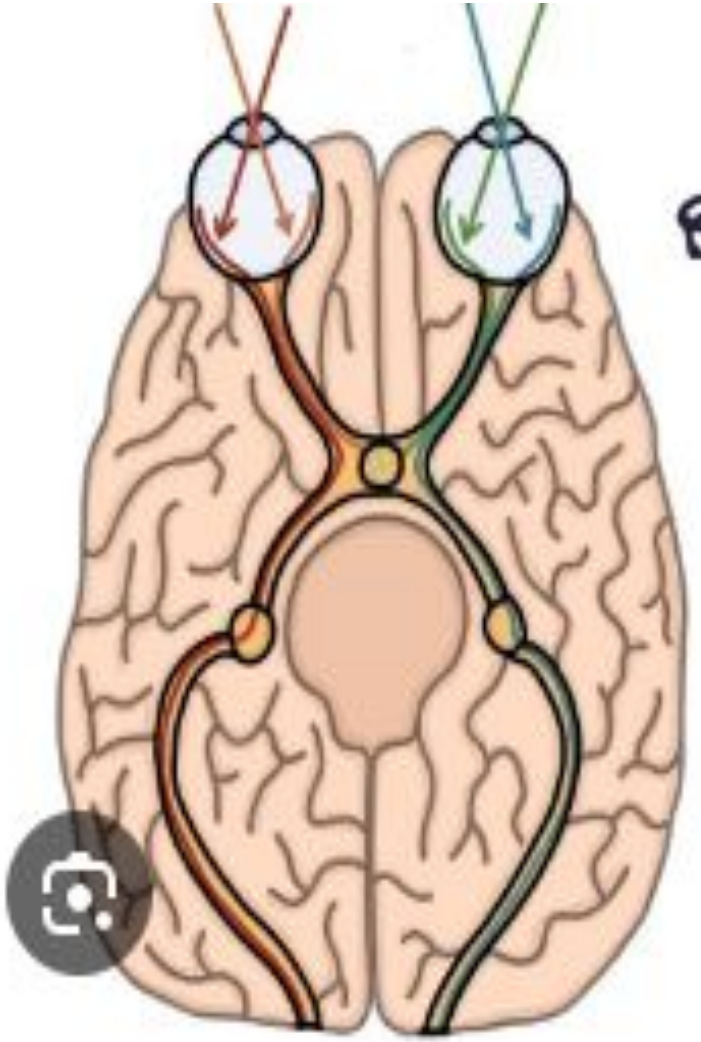
- Prolactin levels are elevated in 30-60% of patients with acromegaly
- High levels of prolactin can result from co secretion of GH and prolactin by pituitary tumor
- Prolactin and growth hormone represent groups of hormones that share similar chemical structures. Both hormones are regulated by similar signaling pathways involving cell growth and hormone secretion

Question 7

How would you relate his visual field defects?

Answer 7

- Pressure on optic chiasm due to pituitary tumor leads to temporal hemianopia visual field defect
- This defect is described as loss of outer half of the vision in both the eyes



BITEMPORAL HEMIANOPIA



LOSS OF VISION
OUTER HALF

Question 8

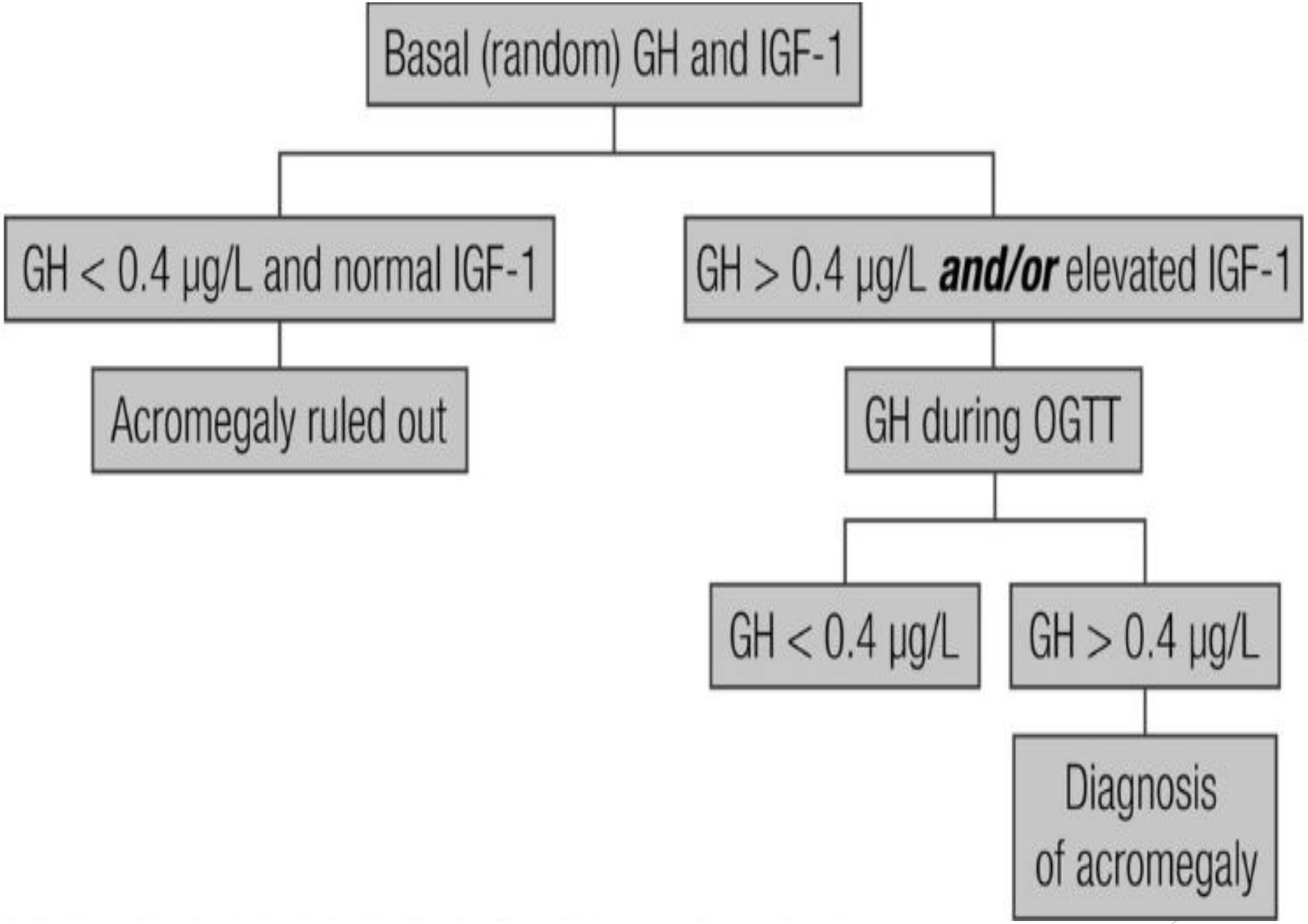
What is the most important blood test in this case?

Answer 8

Growth hormone suppression test

- This is the best method for confirming an acromegaly diagnosis. During this test, your GH blood level is measured both before and after you drink a preparation of sugar (glucose).

- In people who don't have acromegaly, the glucose drink typically causes the GH level to fall


Basal (random) GH and IGF-1
GH < 0.4 µg/L and normal IGF-1
GH > 0.4 µg/L and/or elevated IGF-1
Acromegaly ruled out
GH during OGTT
GH < 0.4 µg/L
GH > 0.4 µg/L
Diagnosis of acromegaly

Question 9

What is the imaging technique of choice in this case?

Answer 9

- Magnetic resonance imaging (MRI)
- MRI help pin point the location and size of a tumor on your pituitary gland

Question 10

What will be the radiographic finding in X Ray of foot/hand/face?

Answer 10

X Ray Hand and Feet

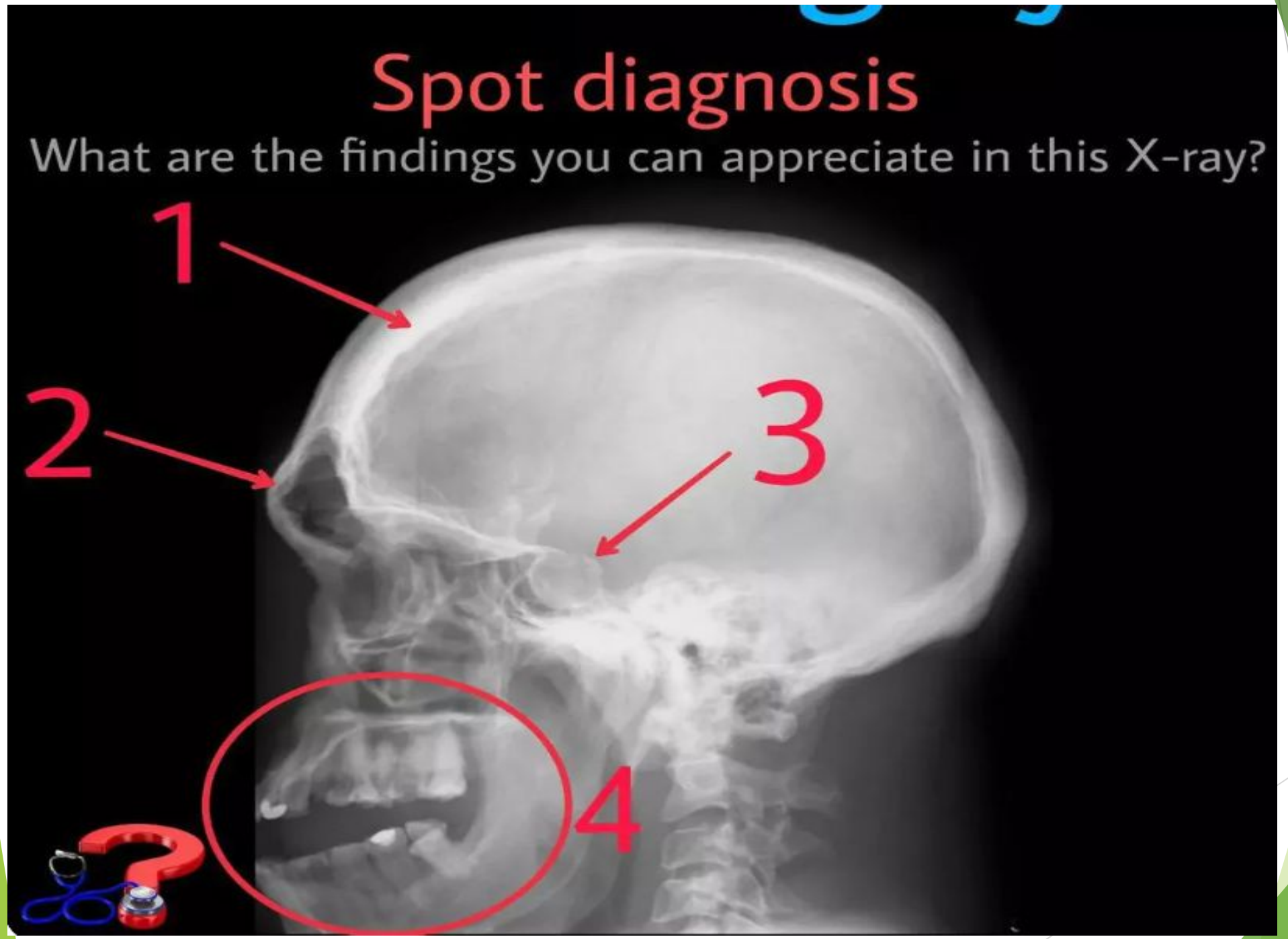
- Widening of the bases of distal phalanges
- Metacarpal osteophytes on radial aspect (metacarpal hooks)



Answer 10

X Ray of Face

- Calvarial (calvarium..bones that protect brain) thickening particularly of inner table
- Frontal bossing and enlarged paranasal sinuses
- Enlarged sella turcica
- Enlarged mandible resulting in prognathism (bulging out of lower jaw/mandible) and gaps between teeth


Spot diagnosis
What are the findings you can appreciate in this X-ray?
1
2
3
4


Thank You!